



Classifying fragments of terracotta warriors using template-based partial matching

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Abstract Large numbers of Terracotta Warriors fragments are mixed up when they are excavated, which calls the need for classifying them in order to avoid one to all matches and improve the efficiency of reassembly. In this paper, we propose a novel template-based classification algorithm for Terracotta Warriors fragments, which transforms the problem of classification into the problem of partial shape matching. Firstly, a few distinct regions, which indicate categories of the fragments, are selected as templates from collections of fragments. Secondly, a novel partial matching procedure is proposed to find whether a fragment have a sub-region that is similar to one of the templates. If there exists a sub-region on the fragment that is similar to one of the templates, this fragment is considered to belong to the category that the matching template represents. The partial matching procedure consists of a coarse matching and a fine matching, which utilizes the normal distribution descriptor and the modified point feature histogram descriptor, respectively. The normal distribution descriptor characterizes the structure of a Euclidean-sphere region and the modified point feature histogram descriptor characterizes the structure of a geodesic-disk region. Experiments have been conducted on Terracotta Warriors fragments that are scanned from real sites to verify the effectiveness of our method. Comparing with other partial matching methods, more accurate classification results are achieved.

Keywords Terracotta warriors · Fragments classification · Partial matching · Point feature histogram

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1 Introduction

Computer-aided cultural-heritage restoration technologies have developed greatly in recent years. Proper matching parts can be found among large numbers of fragments through computational methods, which assist greatly in reassembly tasks. According to the survey of computer methods in reconstruction of 3D archeological pottery objects [25], the computer assists in reconstructing and reassembling archeological fragments in a two-stage process involving fragments classification and reassembly. Classifying methods usually depend on the texture and profile information. Reassembling methods include fracture-region based algorithms [12] and template-based algorithms [36–38]. Fracture-region based approaches exploit similarities in local fracture geometries, whilst template-based algorithms compose fragmented pieces based on the best matches to a full model.

Terracotta Warriors are a collection of terracotta sculptures, which are famous world-heritage objects. The fragments are piled up when they are excavated in archeological sites, as are shown in Fig. 1. The classification of the fragments is essential. Firstly, efficient classification will assist the management of fragments for archeological experts. Secondly, results of categories are foundations of the following reassembly tasks. Previous classification approaches only depend on the profile and texture information, and thus the accuracy is extremely limited.

Innovative methods for the classification of Terracotta Warriors fragments are needed. In this paper, we propose a template-based partial matching algorithm for classifying fragments of Terracotta Warriors using the normal distribution descriptor (NDD) [22] and the modified point feature histogram (PFH) [27]. The mathematical description of 3D shape partial matching is described as follows: assume that a full shape M and a partial shape F are given, and that the partial shape F is a sub-part of M , which is a surface with boundaries embedded in R^3 . The task is to find correspondences between F and M . A few distinct regions, which lie on intact surfaces and indicate categories of the fragments, are selected as templates to uniquely identify categories.

The contributions are listed as follows:

- 1) An innovative partial matching method is used for the classification of Terracotta Warriors fragments, which transforms the classification problem into partial shape matching with the assistance of templates.



Fig. 1 Large collections of the Terracotta Warriors fragments in archaeological site

- 2) A modified point feature histogram (PFH) descriptor is proposed to match fragments with templates, which utilizes geodesic disks with equal radii for matching.
- 3) Comparing with other local shape descriptor-based partial matching methods, our algorithm shows advantages in dealing with smooth fragments.

The remainder of the paper is arranged as follows. Section 2 introduces related works of classification of 3D archeological fragments and the common partial matching methods. Section 3 illustrates the overview of our algorithm. Section 4 introduces the descriptors used for matching. Matching procedures are given in Section 5. Section 6 shows experiments and comparisons. Finally, some conclusions are given in section 7.

2 Related work

In this section, we illustrate related work of the classification of archeological fragments and 3D shape partial matching.

2.1 Classification of archeological fragments

The classification of large amounts of mixed fragments is essential for reassembly, which avoids one-to-all matches and reduces the complexity of reassembly greatly. There are few research works of the classification of archeological fragments.

A bottom-up strategy [30] was proposed to classify fragments using the profile information. The profile section is the section of the fragment in the direction of the rotational axis. The profile section is segmented into its primitives and relations among these primitives, which form a descriptive language to classify the fragments. A method to classify fragments of archeological potteries [17] was proposed using a two-dimensional profile. The two-dimensional profile is formulated as functions that segment the full profile into relevant sub-parts for classification.

A color-specification technique [16] was proposed on the basis that the spectral reflectance of archeological fragments varies slowly. They obtain accurate colorimetric information using a calibrated acquisition system. A new feature descriptor [31] was proposed based on total-variation geometry combined with the well-known SIFT feature, to classify archeological ceramic fragments using texture and color descriptors. In order to assemble fragments automatically with global consistency, two modifications of the traditional relaxation scheme [6] were proposed to decrease the error accumulation. Another novel algorithm [26] was proposed, which relies on intersections of RGB colors among archeological fragments, and extractions of texture features from fragments. By classifying color features and Euclidean distances between texture features, their algorithm can classify the fragments with high accuracy.

The approaches above mainly utilize the profile information or the texture information. However, large numbers of archeological fragments have similar textures and color information, which poses challenges to texture-based classification methods. To make it even worse, most of the fragments are worn and the profile information is extremely difficult to extract, which also leads failure. Therefore, more robust and effective classification methods are needed which employ the geometry information, combined with profile and texture information.

2.2 3D shape partial matching

Among lots of 3D shape partial matching methods, the structural decompositions of object shapes and graph-matching techniques are utilized to build correspondences between nodes representing the parts. Bespalov et al. [2] proposed an efficient approach to the hierarchical matching of 3D models using a scale-space feature decomposition, rooted tree representation of these feature decompositions and dynamic programming matching of vertex-labeled graphs. Biasotti et al. [3] used structural descriptors for sub-part correspondences of 3D shapes. Cornea et al. in [7] presented a many-to-many matching algorithm using curve skeletons of 3D shapes. Since curve skeletons incorporate the notion of parts or components, they can be utilized to conduct partial matching. Suzuki et al. [33] proposed a partial shape matching technique by decomposing 3D models into parts, and the rotation-invariant shape descriptors are extracted as histograms for matching. Ferreira et al. [9] proposed a collection-aware decomposition approach combined with a shape thesaurus and inverted indexes to describe and retrieve 3D models. Bronstein et al. [5] proposed a definition of the significance of a shape part, and the discriminative power with respect to a given shape database is computed to conduct partial matching. These graph-based matching methods make partial matching easier. However, they are limited since only the topology is taken into account, and they mainly suit CAD environments.

Some approaches are based on local shape descriptors. The representative local shape descriptors include spin image [15], snapshot descriptors [21], local spherical harmonics descriptors [18], Laplace-Beltrami descriptors [8, 11, 13, 19], and heat kernel signature descriptors [32]. A survey [20] analyzed these descriptors and many of them have been used for partial matching. Spin image descriptor [15] is widely used in many shape registration tasks. In [35], spin images are used to do feature extraction and skull matching by generating a 2D description for a set of sample oriented points. Heat kernel signatures (HKS) [32] proved to be an informative and isometric invariant descriptor. In [36], HKS were used for feature extraction and registration between skull fragments and template skulls. In [37, 38], the intrinsic shape signatures (ISS) [39] and signatures of histograms of orientations (SHOT) [34] were used to extract and match features for the reassembly of 3D fragments.

Other shape descriptor based methods analyze the properties of a larger portion of objects, such as a sub-part of their surface. Gal and Cohen-Or [24] proposed a curvature-based salient shape feature extraction method and used the feature for partial matching. However, their approach may be inadequate when the shape surfaces are too complex to extract salient shape features. Attene et al. [1] proposed a Fast Reject schema by exploiting the incremental nature of a class of local shape descriptors, which highly reduced the partial matching time. They tested the schema on three different descriptors and show accurate matching results. Arik and Ayellet [14] integrated the feature-point similarity and the segment similarity, and adopted a probabilistic framework to enhance the certainty of the feature-point similarity. However, their method relies on the segmentation of shape, and the results are not accurate when dealing with complex shapes.

3 Overview

The ambition of this paper is to classify the archeological fragments into particular categories for the convenience of governing the data and assisting the following reassembly tasks.

Terracotta Warriors fragments have arbitrary geometry and lack global structure. Nevertheless, there are locally unique regions on intact surfaces of the fragments, which can indicate the category of the fragment. These distinct regions provide significant clues for the classification.

An innovative template-based partial matching algorithm is proposed and the template refers to locally distinct regions, which can uniquely define a category. If there is a subregion on the fragment that is quite similar to one of the templates, the fragment is considered to belong to the category which the matching template represents. Given any kinds of templates, we can find whether a fragment contains this pattern or not. Different from extracting salient points of the fragments, we utilize the structure of neighboring regions of a point, which contains more information.

The framework of the proposed algorithm is illustrated in Fig. 2.

Firstly, the dataset is constructed by scanned Terracotta Warriors fragments, and a few templates are selected from the collections of fragments according to expert knowledge. These templates are locally unique regions, and each of them only exists in one category. The largest enclosed geodesic disks of the templates are extracted to represent them.

Secondly, the fragment is matched with each template using a two-step matching procedure which contains the coarse matching and the fine matching. The coarse matching is to match normal distribution descriptors (NDD) which are computed for each point of the fragment to obtain candidate points. The fine matching is to match the modified point feature histograms (PFH) of candidate points.

Thirdly, the category of a fragment is obtained if the correlation coefficient of PFH is larger than a given threshold. These procedures are repeated for all the fragments until they are all classified.

4 Descriptors

4.1 Normal distribution descriptor

Michael et al. proposed normal distribution descriptor (NDD) [22] to pick the starting point. NDD captures the distribution of surrounding normals compared with the normal of the reference point, which is a local descriptor. NDD is used for coarse matching to reduce the complexity of following steps since the calculation of NDD is simple and fast. Firstly, points that lie inside a sphere with a radius around the reference point are considered surrounding points. Secondly, dot products between normals of the surrounding points and the normal of

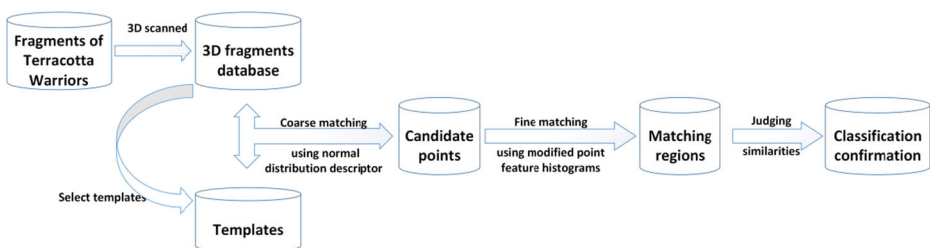


Fig. 2 The pipeline of the proposed algorithm. The dataset is constructed by scanned Terracotta Warriors fragments. Coarse matching and fine matching are conducted using NDD and the modified PFH, respectively. Similarities are judged by the correlation coefficient of PFH

the reference point are computed, which range in $[-1, 1]$. The range is uniformly divided into 10 bins. It should be noted that the range of $[-1, 1]$ is reduced to $[0, 1]$ in our experiments since pointing to the inside of the 3D surface is meaningless. Lastly, NDD is obtained by calculating the percentage of points which the corresponding dot product fell in the respective value range. Fig. 3 shows an example of NDD.

4.2 Modified point feature histogram

Point feature histogram (PFH) [27] was proposed for aligning point cloud views, which was formulated to encode k -neighborhood geometrical properties of a point by generalizing the mean curvature around the point using a multi-dimensional histogram of values. PFH was used to label points accurately in a 3D point cloud [28]. They characterized 3D geometric primitives of interest and described methods for obtaining discriminating features used in a machine learning algorithm. PFH was modified into fast point feature histograms (FPFH) [29], which retain most of the discriminative power of PFH and drastically reduce the computation time.

PFH was originally defined as follows. Firstly, for each point p , all of its neighboring points enclosed in the sphere of a given radius r are selected. If the surface normal n_p at the point is missing, it will be approximated by the normal of the best-fit plane to the neighborhood surface using principal component analysis. Secondly, for every pair of points p_i and p_j ($i \neq j, j < i$) in the k -neighborhood of p , a source point p_s and a target point p_t are selected, with the source being the one having the smaller angle between the associated normal and the line connecting the points. In addition, the Darboux frame with the origin at the source point is defined as shown in Fig. 4, where $u = n_s$, $v = (p_t - p_s) \times u$, and $w = u \times v$. Thirdly, the relationship between p_s and p_t are defined using the following features:

$$\alpha = v \cdot n_t, \phi = u \cdot \frac{(p_t - p_s)}{d}, \theta = \arctan(w \cdot n_t, u \cdot n_t), \quad (1)$$

where d is the Euclidean distance between p_s and p_t , $d = \|p_t - p_s\|_2$. These three features are computed for each pair of points in the k -neighborhood. Lastly, by dividing the feature value range into b subdivisions, b^3 bins are formed. The percentages of pairs that have the same category for all three features are stored in corresponding bins, which constitute PFH.

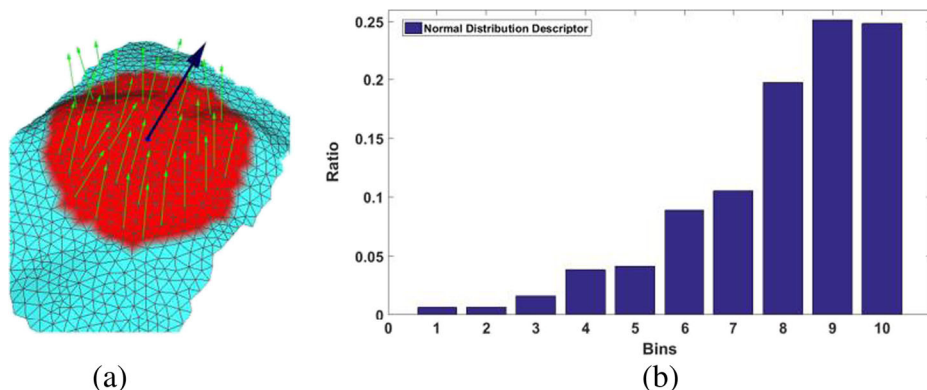


Fig. 3 An example of NDD. The value of NDD ranges in $[0, 1]$, and 10 bins are used

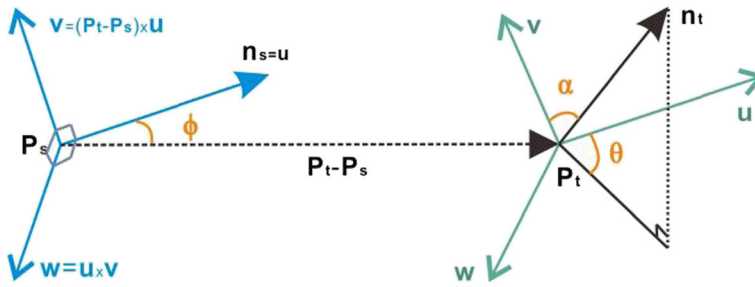


Fig. 4 The Darboux frame of the two points p_s and p_t

PFH uses a higher dimensionality for characterizing the local geometry around a point, and PFH is robust in the presence of outliers and invariant to the position, orientation and sampling densities of the point cloud. In this paper, PFH is modified to assist in partial matching tasks.

In the original calculation of PFH, the neighboring points are selected if they are enclosed in the sphere of a given radius r . However, this may lead to ambiguities in applications for partial matching. The use of Euclidean distance on triangular meshes ignores the connectivity information among points. In addition, the complexity is extremely high when there is a large amount of points within the region. Aiming at these two aspects, we made two modifications to the original PFH.

Firstly, the geodesic-disk region around a point is used rather than the Euclidean-sphere region. Geodesic distance is used to compute neighboring regions instead of Euclidean distance. This neighboring region is called the geodesic disk $GD(p, r)$ with the center p and radius r . Using geodesic distance reserves the connective relationship of the points, and preserves more information of the shape.

Secondly, relationships between p and other points within $GD(p, r)$ are computed instead of all pairs within the disk. In our matching task, the fragments are scanned in a high resolution and have a large amount of points. Our modification greatly decreases the computational complexity. By computing relations from other disk points to the center point, the three features α, ϕ, θ for each pair are obtained as well, which characterizes the geodesic-disk region.

The modified PFH is discriminative since each point has unique neighboring regions. We calculate the modified PFH for the templates used in this paper, which are shown in Fig. 5. The modified PFH is also resolution invariant. We subdivided the torso template and calculated the modified PFH, which is very similar to the original modified PFH, as shown in Fig. 6.

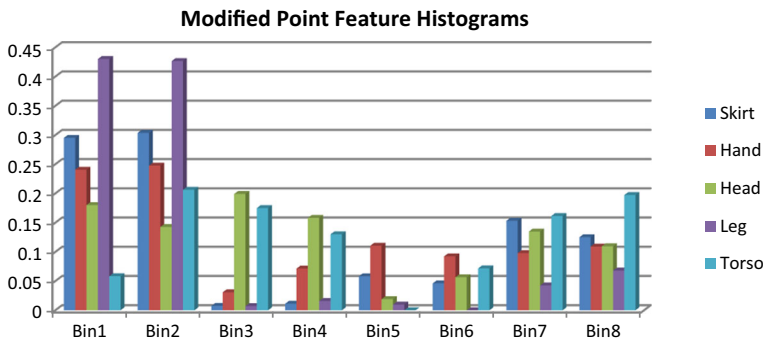


Fig. 5 The modified PFH descriptors of geodesic disks of the templates. Bars with the same color represent the descriptor of each template. Bars in each bin have large variances, which demonstrate the distinctness of the modified PFH

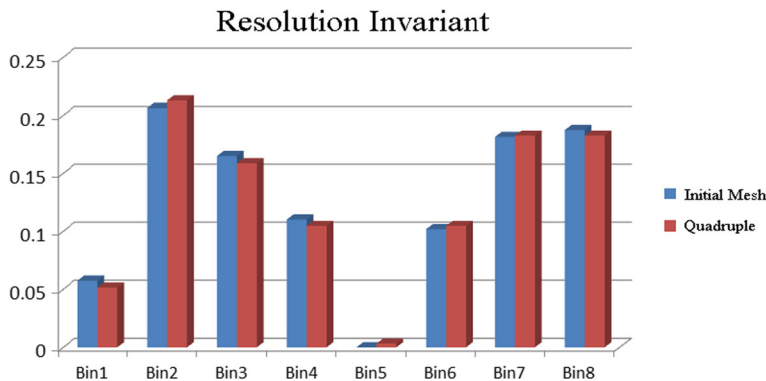


Fig. 6 Resolution invariance. The modified PFH descriptors of templates with different resolutions are quite similar

5 Matching procedures

In this section, we present the template selection procedure firstly, and the two-step matching procedures are then explained in detail.

5.1 Template selection

Terracotta Warriors have various expressions, postures and costumes in intricate geometric details. Fragments that belong to one category can be reassembled firstly to improve the efficiency and accuracy of virtual reassembly. By observing large numbers of fragments, we find that partial surfaces that belong to the same category have similar local characteristics and demonstrate some regularity. Therefore, distinct regions are selected, which uniquely characterize the fragments. When a sub-region of a given fragment is matched with one of the templates, this fragment will be classified into the template category.

The Terracotta Warriors are classified into five categories: head, torso, hand, skirt and leg, which are represented as C_1 , C_2 , C_3 , C_4 , and C_5 , as is shown in Fig. 7. Templates are selected



Fig. 7 Categories of the Terracotta Warriors, which are Head, Torso, Hand, Skirt and Leg

according to whether they can define the category uniquely. Eight templates are selected according to expert knowledge, which are shown in Fig. 8. In the Head category, eyes, ears and the mouth are selected as the templates. In the Torso category, the nail on the suit is selected. In the Hand category, the fist is selected. In the Skirt category, the skirt edge is selected. In the Leg category, the region on the leg is selected.

5.2 Geodesic disk computation

Templates always contain arbitrary boundaries since they are selected manually. Comparing surface regions on the fragment with consistent boundaries of the templates is impractical. Previous feature-point extraction and matching approaches may fail if the matching regions have few distinct feature points or the descriptors are not distinctive. Therefore, we compute the largest enclosed geodesic disk for templates in order to make the matching comparable.

Magdalene and Marc [4] defined the geodesic disk in a simple polygon, and the geodesic disk of a 3D manifold shape is defined in this work. Let S be a 3D manifold surface of n vertices. A geodesic disk $GD(p, r)$ with center p and radius r is the set of points of S that have a geodesic distance no larger than r from p . On a discrete polygonal mesh, the geodesic distance is the length of the shortest polygonal path between source and target points.

The largest enclosed geodesic disk has the largest geodesic radius of the template surface, and the radius will be used further. The computation of the largest enclosed geodesic disk has the following steps. Firstly, all boundary points of the template surface are taken as source points. Secondly, each non-boundary point is taken as the target point, and the shortest geodesic distances to the sources are computed. Thirdly, among these shortest geodesic distances, the largest distance is adopted as the radius of the largest enclosed geodesic disk, and the corresponding target point is adopted as the center of the largest geodesic enclosed disk. The largest enclosed geodesic disks of the templates are shown in Fig. 8.

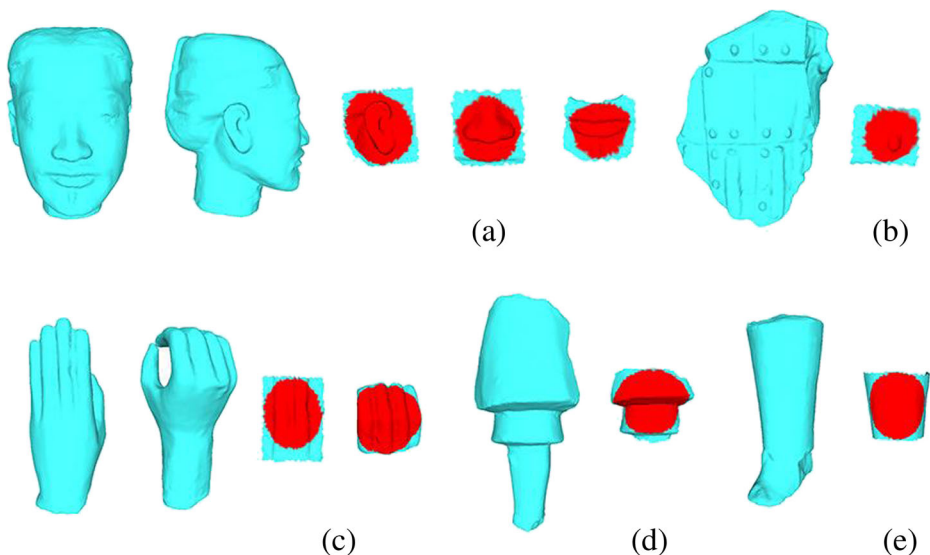


Fig. 8 Selected templates and their geodesic disks. (a) Templates of the Head category. (b) Templates of the Torso category. (c) Templates of the Hand category. (d) Templates of the Skirt category. (e) Templates of the Leg category

5.3 NDD-based coarse matching

The corresponding correlations are adopted to evaluate the similarity between two regions, where each region is represented as feature vectors. The corresponding correlations measure the degree of correlation of two sets of variables, which are widely used in the field of natural science. The correlation coefficient of two sets of variables is defined as follows:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}, \quad (2)$$

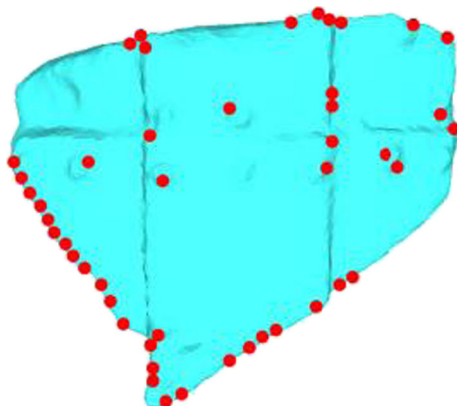
where $X = \{x_1, x_2, \dots, x_n\}$ and $Y = \{y_1, y_2, \dots, y_n\}$ are two sets of variables, $\bar{X}$ and $\bar{Y}$ are the average values. The correlation coefficient is adopted to measure the linear approximity between the two sets of variables, which ranges within the interval $[-1, 1]$. The absolute value of the correlation coefficient is commonly used which ranges in $[0, 1]$. A greater value indicates stronger correlation between the compared features. A correlation coefficient of one means the two vectors are equal. If the value is above 0.6, the two sets of variables will be strongly relevant.

The similarity of two regions is computed firstly using the correlation coefficient between respective NDD. Firstly, we calculate NDD for the central point p of the template's geodesic disk. Euclidean distances from p to boundary points are calculated and the shortest distance is chosen as the sphere radius of the Euclidean sphere. Secondly, for each point of the fragment, NDD is computed with equal radii. Thirdly, we compute the correlation coefficient between NDD of the template and NDD of each point on the fragment. Lastly, N similar points are selected according to their corresponding correlation coefficients. In our experiments, if the correlation coefficient is above 0.7, the point will be considered as the candidate point for further matching. The threshold is set to ensure the covering of actual matching points. 104 candidate points of one Terracotta Warriors fragment are obtained for the Torso template, as is shown in Fig. 9.

5.4 Modified-PFH-based fine matching

If a fragment belongs to one category, this fragment should contain at least one region whose local structure is similar to the corresponding template. Since each template region is

Fig. 9 The remaining candidate points after NDD-based matching. These points will be used for the next matching procedure



represented by a geodesic disk with a fixed radius, geodesic disks with equal radii are extracted for candidate points on the fragment.

During the computing of geodesic disks, calculations regarding geodesic distances from one point to all other points are required. However, geodesic disks merely contain points whose distance to the central point is less than the radius. An optimized geodesic disk computing method is proposed to reduce the computational complexity. The geodesic distance between two corresponding points of the partial shape and the full shape is consistent, and the Euclidean distance between two surface points is no larger than their geodesic distance. Therefore, a particular Euclidean sphere region centered at one point is selected to compute the geodesic disk. The time complexity is greatly reduced using this optimized method. The procedure of the optimized method is shown in Fig. 10. Some geodesic disks of the candidate points are shown in Fig. 11. In this case, the computation time for the 104 geodesic disks is 13.637 s using the improved method, whereas 1347.155 s are needed using the traditional method.

The modified PFH descriptors are computed for geodesic disks whose centers are the candidate points, which will be further compared with the modified PFH of the template region. We segment α, ϕ, θ into two bins respectively, and obtain a vector with eight dimensions. The matching of the vectors is conducted using the correlation coefficient as well. In our experiments, if the correlation coefficient of two disks is above 0.6, the two disks will be considered similar. With this threshold, six similar disks among the geodesic disks of candidate points are found, which are shown in Table 1. Therefore, we came to the conclusion that this fragment belongs to the Torso category.

6 Experimental results and analysis

6.1 Experiments

Experiments are conducted on 113 Terracotta Warriors fragments which are classified by archaeologists. The results of classification are considered the ground truth. The fragments are preprocessed to remove noise. Each fragment is compared with the templates to determine its category, and the results are used to verify our algorithm. One fragment

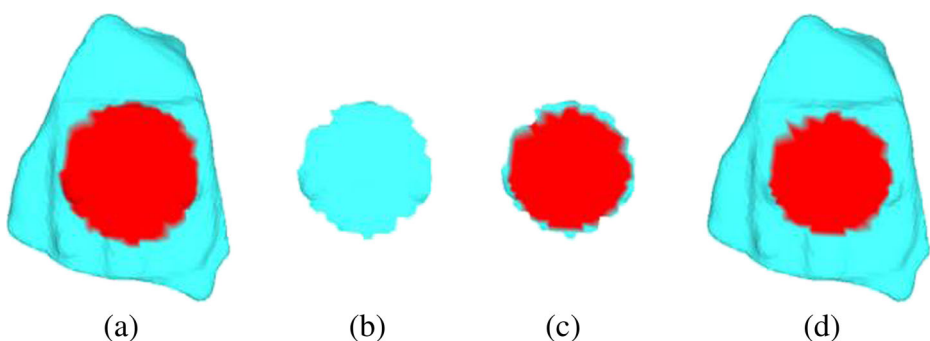


Fig. 10 The optimized geodesic disk computing method. (a) The Euclidean sphere region (red region) of the full shape. (b) The extracted Euclidean sphere region. (c) Geodesic disk computed of the Euclidean sphere region. (d) The geodesic disk region of the full model through back mapping

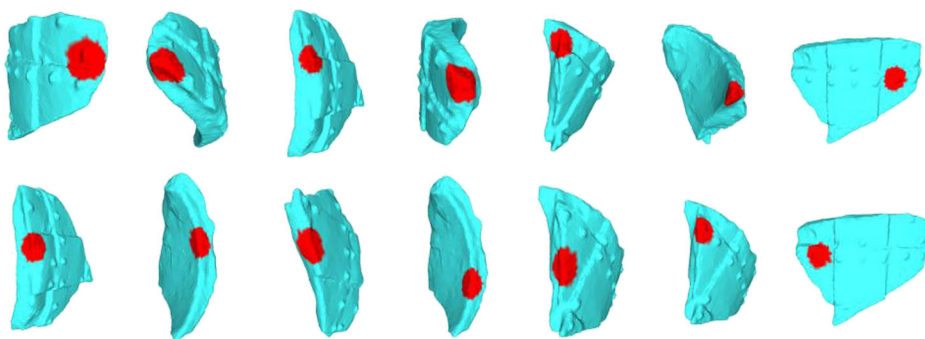


Fig. 11 Geodesic disks whose centers are the candidate points on the fragment

may belong to several categories and any of categories are considered a correct matching. Eight template regions are selected according to expert knowledge. These templates are sorted by their radii of their corresponding largest enclosed geodesic disks. Their radii are shown in Table 2.

All fragments are sorted by their approximate diameters in order to improve efficiency, and the sorting results are illustrated in Fig. 12. We first select the fragment whose approximate diameter is the smallest and compare it with the template whose corresponding radius of the geodesic disk is the smallest. If the fragment cannot be matched with this template, a larger template will be used. During the comparison, if the approximate diameter is less than the diameter of the template's geodesic disk, the fragment will not be compared with this template.

Our algorithm is implemented on a 64-bit PC with an Intel Xeon 2.13GHz CPU and 6 GB of memory. The geodesic distance is calculated using the MMP algorithm [23]. Ten bins are used for NDD and eight bins are used for the modified PFH. In our experiments, if the correlation coefficient is above 0.7, we consider this point as the candidate point for further matching. If the number of candidate points is not larger than thirty, we will take these points as candidate points. If the number of candidate points is larger than thirty, we will take the first thirty similar points as candidate points. This improves the matching efficiency, since we do not need to find all similar sub-regions on the fragment. If there exists at least one geodesic-disk region on the

Table 1 Partial matching results of Torso template for fragment named G10–40–37

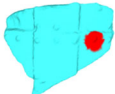
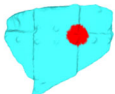
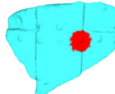
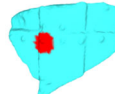
Reference template					
Sub-regions					
Similarity	0.818450	0.798308	0.708839	0.652514	0.639249

Table 2 Radii of different largest enclosed geodesic disks of the templates

Category	Template surface	Number of points in template	Number of points in geodesic disk	Radius of the geodesic disk
C ₄	Skirt	1687	1064	118.774692
C ₁	Ears	1450	1150	59.705657
C ₅	Leg	2339	1130	56.873776
C ₃	Hand-1	2446	1746	47.690055
C ₁	Nose	842	409	41.077855
C ₃	Hand-2	1115	715	33.988004
C ₁	Mouth	639	369	32.298342
C ₂	Nail	405	223	18.900479

fragment, whose correlation coefficient with the template is above 0.6, we will consider this fragment to belong to the category that the template represents.

Take a Torso fragment as an example. Twenty points are obtained whose neighboring disk regions have larger correlation coefficients with the Torso template, as are shown in Table 3. This fragment has 20 sub-regions that are similar to the Torso template. Therefore, this fragment is classified into the Torso category.

Our method is invariant with triangulations and rigid transformations. The fragments have different resolutions and different positions compared with the template, as are shown in Table 4. Our method finds the matched geodesic disks and accomplishes the matching task correctly.

During the classification, there exist fragments which belong to two or more categories. This kind of fragment may have a large size and covers several categories. For example, the fragment named G10–22-5 belongs to both the Skirt category and the Leg category. Table 5 and Table 6 give the partial matching results for different templates.

The overall correct ratio is 89.3% for all fragments, and the correct ratio for each category is illustrated in Table 7, which shows that the accuracy of each category achieves a higher degree. This is because of the unique templates we selected and the proposed accurate partial matching algorithm.

A short time is required to accomplish the partial matching task. Time costs are counted for the two matching stages. The NDD-based matching utilizes the normal distribution descriptor which is easy to compute and costs a relatively shorter time. The modified-PFH-based matching utilizes the modified-PFH which needs much time for computing and matching. The average time cost for each kind of category is shown in Table 8. The time costs depend on the number of fragments and the scale of the template. Generally, if the fragment has a small number of vertices and the template also has a small radius, the algorithm will be fast.

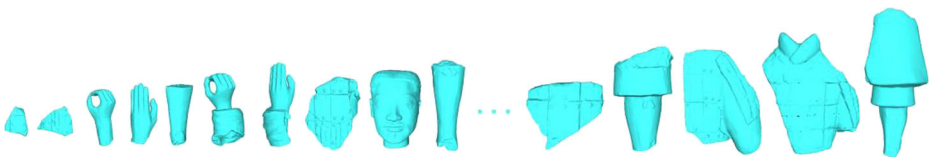
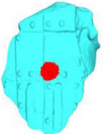
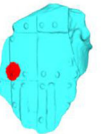
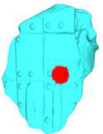
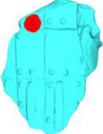
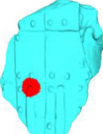
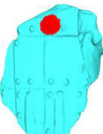
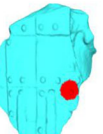
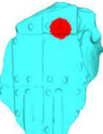
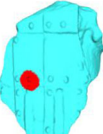
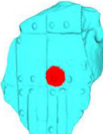


Fig. 12 Sorting results of fragments with different approximate diameters. The fragment with the shortest approximate diameter is matched firstly

Table 3 Partial matching results of Torso template for fragment named G10–41–32. Here we only give the first several similar regions

Reference template					
Sub-regions					
Similarity	0.999774	0.925922	0.908002	0.820424	0.803854
Sub-regions					
Similarity	0.798264	0.787224	0.786966	0.786521	0.785946

6.2 Comparisons and discussions

Comparisons are made with spin image [15] based partial matching method, SHOT [34] based partial matching method and HKS based partial matching method [10, 32]. These methods are typical local shape descriptor-based partial matching methods. The method in [10] positioned each fragment with the assistance of a full Terracotta Warriors model. Therefore, we utilize the three descriptors to accomplish the positioning task. The positioning results indicate the category information.

Comparisons are made on the experimental datasets in this paper. During comparisons, the ISS [39] detector is used to extract feature points from the full Terracotta Warriors model and fragments. The spin image descriptor [15] and the SHOT descriptor

Table 4 Partial matching results of Leg template for fragment named G10–36–19. The fragment has different resolutions with the template region

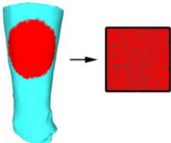
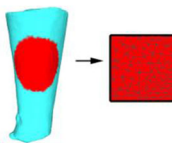
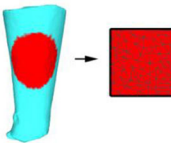
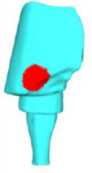
Reference template			
Sub-regions			
Similarity	0.997251	0.997039	0.996951

Table 5 Partial matching results of Leg template for fragment named G10–22-5

Reference template					
Sub-regions					
Similarity	0.998323	0.997424	0.997214	0.996343	0.996119
Sub-regions					
Similarity	0.996022	0.993670	0.993361	0.993194	0.988746

[34] are used for matching. Aiming at HKS-based partial matching method, we used the feature point extraction and matching method in [10]. An example is shown in Fig. 13, which shows the positioning results of the three algorithms. Results of our method are shown in Table 4. The three algorithms all achieved wrong positioning results since the fragment is quite smooth and the feature points have no capacity to differentiate the shape. Whereas, our method utilizes the regions for matching and finds the correct matching positions. Quantitative comparisons are shown in Fig. 14, which demonstrate that our algorithm outperform the three algorithms.

Table 6 Partial matching results of Skirt template for fragment named G10–22-5

Reference template					
Sub-regions					
Similarity	0.980714	0.980193	0.977974	0.977392	0.974039

Table 7 Classification results for each category

Category	Number of fragments	Correct results	Correct ratio
C ₁	8	7	87.5%
C ₂	75	68	90.7%
C ₃	10	8	80%
C ₄	10	9	90%
C ₅	10	9	90%

First of all, incorrect corresponding results are caused for their algorithms since many feature points extracted by ISS and HKS lie on the fracture surface of the fragment. These feature points do not exist on the full shape, which leads to the failure of positioning. On the contrary, our algorithm is more accurate by using the intact surface of the fragment, which maintains the connectivity between points and reserves more information.

Secondly, the discrimination of the extracted feature points is weak when the fragment is smooth. However, the modified PFH of a geodesic-disk region is calculated in our algorithm, which is more discriminative than the point-based descriptors. As is shown in Fig. 14, classification accuracy in C4 and C5 of spin image [15], SHOT [34] and HKS [10] are obviously lower than our method since the fragments in these categories are quite smooth. Therefore, our algorithm achieves better results when dealing with smooth fragments.

Thirdly, their algorithms are sensitive to triangulations, while our method is invariant to triangulations. The results of Table 4 and Fig. 13 demonstrate the effectiveness of our algorithm.

7 Conclusion

The classification of Terracotta Warriors fragments is of high significance since it can help the management of the archaeological data and improve the reassembly efficiency. A two-step matching procedure is proposed in this paper to accomplish the task, which consists of a coarse matching and a fine matching. If one fragment has one sub-region whose correlation coefficient with the template reach a threshold, these two regions will be considered similar and the fragment will be classified into the category the template represents.

Our algorithm utilizes an innovative strategy which classifies 3D archeological fragments using template based partial matching methods, rather than extracting local shape features. The template is rather small compared with the fragments and is to be matched

Table 8 Average time costs of the NDD-based matching and the modified-PFH-based matching for each category

Category	Coarse matching (s)	Fine matching (s)
C ₁	18.645	15.583
C ₂	9.223	13.607
C ₃	15.734	6.345
C ₄	14.316	25.589
C ₅	11.595	10.869

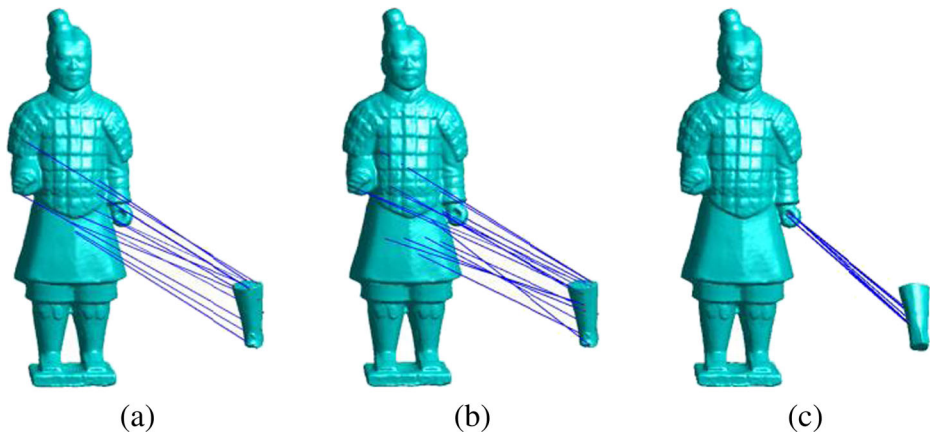


Fig. 13 Comparisons with (a) spin image based method [15], (b) SHOT based method [34] and (c) HKS based method [10]. The three algorithms all achieved wrong positioning results

with a sub-region of the fragment. This part-to-whole matching problem is conducted by restricting the matching area using Euclidean spheres and geodesic disks. The modified PFH descriptor is proposed as the disk descriptor, which describes local features discriminatively.

Experiments are conducted on Terracotta Warriors fragments that are scanned from real sites, and the results show high classification accuracy. Our algorithm is flexible, accurate, and more robust comparing with other local shape descriptor-based partial matching method. Even if the fragment is smooth, our algorithm can also classify it if the smooth region is uniquely defined. Our algorithm can be expanded to any other archeological artifacts if they have unique regions from which their categories can be identified.

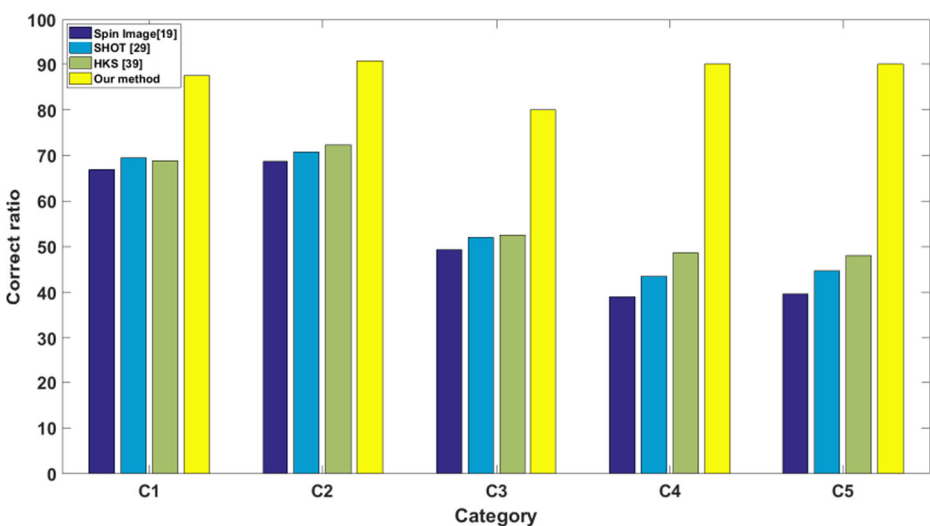


Fig. 14 Classification results of the three algorithms compared with our algorithm

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